Sahaj Bhadja

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Summary

Solutions-oriented engineer with 2+ years translating **customer requirements and quantitative data into practical technical designs**; experienced in calculations, testing, cost tracking, documentation, technical presentations, process improvement, and project handoff.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Partnered with stakeholders to gather, validate, and interpret **operational and technical requirements**, then developed scalable solution options supporting 500+ users while reducing dashboard latency from 1.2 seconds to 350 milliseconds.
- Collected, organized, and analyzed 10,000+ records to validate system designs, implementing reconciliation, discrepancy detection, and reliability checks that sustained 99%+ accuracy across 5,000+ recurring monthly workflows.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Translated customer workflows into **clear technical designs and process options** covering uploads, processing, dashboards, and reporting, serving 150+ users and 10,000+ daily requests with reliable performance.
- Tracked usage and estimated costs across multiple AI solutions, presenting technical trade-offs and incorporating stakeholder feedback while reducing turnaround 60% through tested automation and organized project execution.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Performed **detailed quantitative engineering analysis** across 100,000+ trajectory records, translating research requirements into repeatable simulations, calculations, and practical design recommendations that improved long-horizon planning performance 18%.
- Designed and executed 10,000 system-level simulation tests, analyzing road geometry, directional data, failure conditions, and performance metrics to improve closed-loop stability 15% while documenting experimental findings.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Served as a **technical liaison** with Microsoft and GitHub engineers across regions, gathering requirements, validating designs, and aligning deliverables while improving response times 30% and process efficiency 68%.
- Prepared technical documentation and automated 20+ validation tests across Windows, Linux, and macOS, producing clear results and discrepancy reports while supporting reliable handoff to globally distributed engineering teams.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Worked directly with customers to gather, analyze, and present **end-to-end solution designs** covering booking, pricing, payments, notifications, and operational reporting across 50,000+ records with sub-200-millisecond responses.
- Mapped and documented 12+ technical integration flows into clear implementation proposals, resolving specification discrepancies and coordinating project delivery while reducing manual refund handling 70% through practical automation.

Skills

Application Engineering: Requirements gathering, specification review, technical options, feasibility analysis, technical proposals, project handoff
Analysis: Engineering calculations, data collection, data analysis, percentages, ratios, performance metrics, cost tracking, discrepancy detection
Testing: Test planning, system testing, validation, expected-versus-actual analysis, root-cause analysis, technical reports
Tools: Microsoft Excel, PowerPoint, Microsoft Office, Python, SQL, Tableau, Git, Jira, process documentation
Collaboration: Customer support, technical presentations, cross-functional teamwork, international teams, prioritization, continuous improvement

Projects

CPU Scheduling Simulator Live | *C++, Calculations, Visualization*

- Implemented and evaluated **six scheduling calculations** with interactive controls, Gantt visualization, comprehensive edge-case testing, and dynamic performance metrics across 1,000+ process combinations, reducing average turnaround time 25%.

CaptionCraft Live | *Testing, Data Analysis, Python*

- Designed a **repeatable performance-testing study** across four technical models, 10,000 images, and 100,000 captions, analyzing test data and failure patterns to improve measured accuracy from 81% to 96%.

HMIS Live | *Workflows, Documentation, Monitoring*

- Developed a **customer-focused operational solution** with role-based access, dashboards, appointments, integrations, notifications, and monitoring, supporting 100+ daily workflows while reducing record retrieval time 50% and administration 30%.